

Angle-Based Sitting Posture Recognition System

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Abstract—The followers of sitting postures played a very crucial role in the prevention of musculoskeletal disorders, particularly among the people who are required to sit for long hours. The system evolved in this study is the posture recognition framework that was developed to identify suboptimal sitting positions and alert the user if it holds for a predefined duration. The algorithms designed in the field of machine learning are strictly suited to ensure that the accurate and reliable feedback about various poses of different body types and interference by extraneous disturbances will be obtained. This proposed solution can reap benefits in opening wide avenues for betterment of generally spinal health by integrating productive individuals in offices, educational institutions, and even in home surroundings.

Index Terms—Sitting posture detection, Posture recognition system, Real-time monitoring, musculoskeletal disorders.

I. INTRODUCTION

Posture detection technologies have become essential in addressing health risks related to prolonged sitting, including chronic back pain and spinal misalignment. Current systems offer instant feedback, yet they often struggle with reliability due to algorithmic limitations, body type variability, and environmental noise [1], [2]. While wearable sensors provide detailed posture data, they can be cumbersome, and vision-based systems face challenges like lighting and occlusion issues [3], [4], [5]. This study aims to create a more adaptable posture detection system, providing real-time feedback across diverse settings to enhance user awareness and promote healthier sitting habits.

A. Related Work

Emerging technologies are increasingly focused on addressing the health risks associated with prolonged sitting through posture detection and correction systems, which can be broadly categorized into sensor-based and vision-based approaches, each with its respective advantages and disadvantages. Sensor-based systems rely on wearable technologies, such as accelerometers and gyroscopes, that detect spinal alignment and provide immediate feedback. While beneficial for ergonomic analysis, these systems require multiple sensors, complicating their adaptation for home use. Classification of

human postures typically utilizes machine learning algorithms in conjunction with feature extraction techniques; however, the reliance on fixed features in these algorithms can lead to rigidity and reduced effectiveness in dynamic environments [1]. In contrast, vision-based systems leverage camera setups to detect postural deviations without the need for physical sensors. CNN-based approaches have shown promise in human pose estimation, yet they tend to be computationally intensive, which limits their use in real-time applications [2]. Advanced models like AlphaPose utilize skeletal and depth information for real-time whole-body pose estimation, significantly enhancing accuracy and robustness. However, these systems often face challenges related to varying luminance, viewing angles, and occlusion effects, which can compromise reliability [3] [4].

The need for robust models is further emphasized by the insights provided in a comprehensive survey on deep learning-based 2D human pose estimation [5]. Deep learning models, especially CNNs, have largely driven advancements in human pose estimation, demonstrating excellent performance in various applications [6] [7]. Models such as HigherHRNet have improved scalability for multi-person pose estimation by learning better spatial representations, yet real-time applicability across diverse environments remains a complex challenge [8]. Recent advancements, such as AlphaPose, highlight effective tracking and regional pose estimation for multiple individuals in real time [9] [10]. Additionally, foundational approaches like DeepPose have contributed to the field, achieving promising results despite limitations in handling intricate scenarios [11]. Techniques such as covariance scaled sampling have further refined articulated human motion estimation, enhancing the robustness of these systems [12]. Recent studies emphasize the importance of high-resolution representation learning for improved pose estimation outcomes [13].

Notably, advancements like Vnect have shown the potential for real-time 3D pose estimation using a single RGB camera, further illustrating the rapid evolution of pose estimation technologies [14]. Innovations in real-time multi-person 2D pose estimation, using techniques such as part affinity fields,

underscore ongoing efforts to improve these systems [15]. Despite notable progress in posture detection technologies, achieving fully responsive, real-time recognition in typical settings remains an unsolved problem, necessitating continued research and innovation to bridge this gap and enhance health and usability outcomes.

B. Our Contribution

According to the author's analysis, current literature on posture detection systems suffers from several disadvantages in dealing with real-time angle-based posture recognition that adapts to different body types and user environments. Most approaches that are currently followed are either computationally expensive or rely on sensor-based methods, which can be invasive and less practical for continuous monitoring in typical settings. By this, it develops a real-time posture detection solution on Mediapipe keypoints extracted then combined with an angle-based analytical method for producing effective and timely feedback without intricate hardware configuration requirements.

This system uses single RGB cameras to detect inappropriate sitting postures with an adaptive tolerance mechanism for feeding back corrective information. The given system combines angle-based evaluation along with adaptive thresholding to recognize the right and wrong postures. Specifically,

- Angle-Based Analysis determines key angles such as neck angle, torso orientation, and shoulder deviation that allows the system to detect frequent postural variations such as slouching and forward head postures besides non-equal shoulder configurations.
- Dynamic Tolerance Mechanism adjusts to each user's natural variations in posture and adjusts thresholds in real-time based on recent data to minimize false positives and personalize feedback.

This paper aims at developing an efficient, lightweight posture detection system that provides accurate, unobtrusive, and real-time feedback suitable for both office and home settings.

This paper is organized as follows: Section II, Mathematical Framework, covers basic angular calculations, tolerance adjustment mechanisms, and posture classification. Section III includes methodology by data collection, key point detection using Mediapipe, and angle-based metrics that should be supplemented by real-time feedback mechanisms. Results are presented in Section IV by including accuracy distributions, system performance, and comparison to other models. It also outlines challenges like data diversity, real-time processing, and refining subtle posture variation detection to increase accuracy and robustness. Finally, Section V concludes with a complete discussion on the sitting posture recognition system.

II. MATHEMATICAL FRAMEWORK

The smallest component of the posture detection algorithm consists of determination of the angles formed by the given body keypoints acquired by Mediapipe. The following section is the mathematical principle underlying these computations,

when considering the assessment of posture and provide feedback.

A. Terminology

1) *Angle Calculation*: To measure the posture of the user we measure angles between key joints: neck and torso for example. The angle θ between points (x_1, y_1) and (x_2, y_2) is given by the arctangent function:

$$\theta = \arctan \left(\frac{y_2 - y_1}{x_2 - x_1} \right) \quad (1)$$

That is, this angle of the user's neck to torso orientation is critical in determining these postural deviations.

2) *Dynamic Tolerance Calculation*: Given the susceptibility of the system to posture change as well as user-specific variation, it updates dynamically using the recent measurement the tolerance level. Tolerance update over time may be represented as:

$$\text{Tolerance} = \mu + \sigma \quad (2)$$

where μ is the average of recent angular measurements, and σ is the standard deviation.

3) *Mean and Standard Deviation*: The average value μ of the angles is given as

$$\mu = \frac{1}{N} \sum_{i=1}^N \theta_i \quad (3)$$

where N is the number of recent measurements, and θ_i is the i -th angle.

The standard deviation (σ) is defined as:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (\theta_i - \mu)^2} \quad (4)$$

It tends to make the system adjust tolerability accordingly as the fluctuations occur with movements of a user.

4) *Posture Classification*: With the calculation of the tolerance and angles, it will categorize whether it is "good" or "bad" according to if angle θ falls into a predefined tolerance range:

$$\text{Posture} = \begin{cases} \text{Good} & \text{if } |\theta - \theta_{\text{ideal}}| \leq \text{Tolerance} \\ \text{Bad} & \text{if } |\theta - \theta_{\text{ideal}}| > \text{Tolerance} \end{cases} \quad (5)$$

where θ_{ideal} indicates the optimal angle giving rise to the appropriate posture. It is, in effect, a manner of providing real-time feedback to the user.

5) *Real-Time Feedback Mechanism*: The system monitors deviations from θ_{ideal} , so that the user is quickly alerted when a suboptimal position is maintained for an appropriately large time. To activate this mechanism, one could employ the timing threshold T :

$$\text{Alert} = \begin{cases} \text{Trigger} & \text{if } t > T \\ \text{No Alert} & \text{if } t \leq T \end{cases} \quad (6)$$

where t is the time by which incorrect posture is detected, and T is the threshold duration for which the body can be in that position. This value can be changed according to the need of the environment the system is installed in.

B. Theoretical Models

A theoretical ground for effective construction of sitting posture detection relies on a set of ideas rooted in biomechanics, computer vision, and mechanisms of feedback adaptation. Such basic notions combined bring about real-time recognition of one's posture, answering actual user's needs.

1) *Biomechanical Principles of Human Posture*:: The system design has reduced pressure against human musculoskeletal structure through tension reduction at the neck, shoulder, and torso using biomechanical principles, which define optimal possible angles for both muscle and joint relaxation. Therefore, posture-related diseases' risk decrease in critical angle calculations conducted during this study leading to supplying time-wise feedback rules and standards of ergonomics to the system.

2) *Pose Estimation for Keypoint Detection*:: The system relies on pose estimation in determining the significant body landmarks within video frames through neck, shoulders, and hips. A CNN or any similar deep learning architecture is processed to carry out pose estimation algorithms in processing image data for constructing spatial relationships between joints and mapping them into 2D coordinates from which angles are computed for posture assessment.

3) *Adaptive Thresholding and Real-Time Feedback Mechanism*:: This method of adaptive thresholding varies the tolerance levels, with recent measurements taken in mind along with natural variations amongst users' postures. For differentiating between minor and major deviations, it is utilizing statistical models of the mean and standard deviation such that false alarms are avoided, but monitoring is highly effective. This immediate feedback mechanism is based on the theory of behavior modification by which the user can adjust posture as soon as the major deviation is reported.

4) *Time-Based Behavioral Correction*:: A time-based alert system is premised on the premises of the reinforcement theory of behavioral psychology. The system emulates a form of negative reinforcement by alerting the users at such long time frames of suboptimal posture and even before forcing them to begin experiencing pain or strain due to their posture Misalignment. This is founded on the belief that constant feedback over time will enable users to learn good posture habits indirectly leading to a reduction of improper posture events.

III. METHODOLOGY

A. System Overview

The sitting posture detection system utilized a webcam to scan through live video feeds, which later would apply Mediapipe to extract keypoints and further analyze using angular measurements. This system scans improper postures, which depend upon neck tilt, torso positioning, and shoulder displacement. The design was to be non-intrusive yet still applied in several settings-homes or work settings. Fig. 1 describes the workflow of the system and signifies each process in a procedural fashion.

B. Data Collection

To make the system stronger, data was captured in multiple subjects with various age ranges, body types, and sitting conditions. Even the setting was deployed under multiple lighting and camera angle exposures that would look at actual exposures. Frames have been annotated carefully as being in "good" posture or "bad" according to standard ergonomic conditions. Thus, aggregate 5000-annotated frames have used to train and test while obtaining a balanced dataset towards the validation of models.

C. Keypoint Detection

The Mediapipe pose estimation module provides 33 keypoints for rendering the different body landmarks. The following keypoints have been considered for this posture detection system:

- Neck (Point 0)
- Left Shoulder (Point 11)
- Right Shoulder (Point 12)
- Left Hip (Point 23)
- Right Hip (Point 24)

The keypoints are actually utilized to calculate angles of the neck and torso beside checking the alignment of shoulders. When positional data for keypoints is readily available, then this system may measure angles and deviation from ideal posture. Fig. 2. shows all the body keypoints that can be extracted using Mediapipe model.

D. Angle Analysis

Using keypoints found, the system calculates angles needed for assessment of posture:

- Neck Inclination: The measurement done between the neck and the upper torso determines whether the user is in posture upright or otherwise. Values are assumed to lie in the range of 10-15 degrees along the torso line; such values suggest a forward head posture.
- Torso Inclination: Angle between upper and lower back, calculated via the hip and shoulder joints. This is a means of detecting slouching. Ideal is considered a slight backward tilt wherein deviations greater than 5-10 degrees is considered as bad posture.

E. Shoulder Offset

Shoulders are measured as the proper vertical positioning of the left shoulder and the right shoulder. A difference of more than a few centimeters between the two shoulders would indicate that there is asymmetric placement that could manifest in slumping or slouching to one side. The algorithm calculates an offset in centimeters across which an alert will be generated.

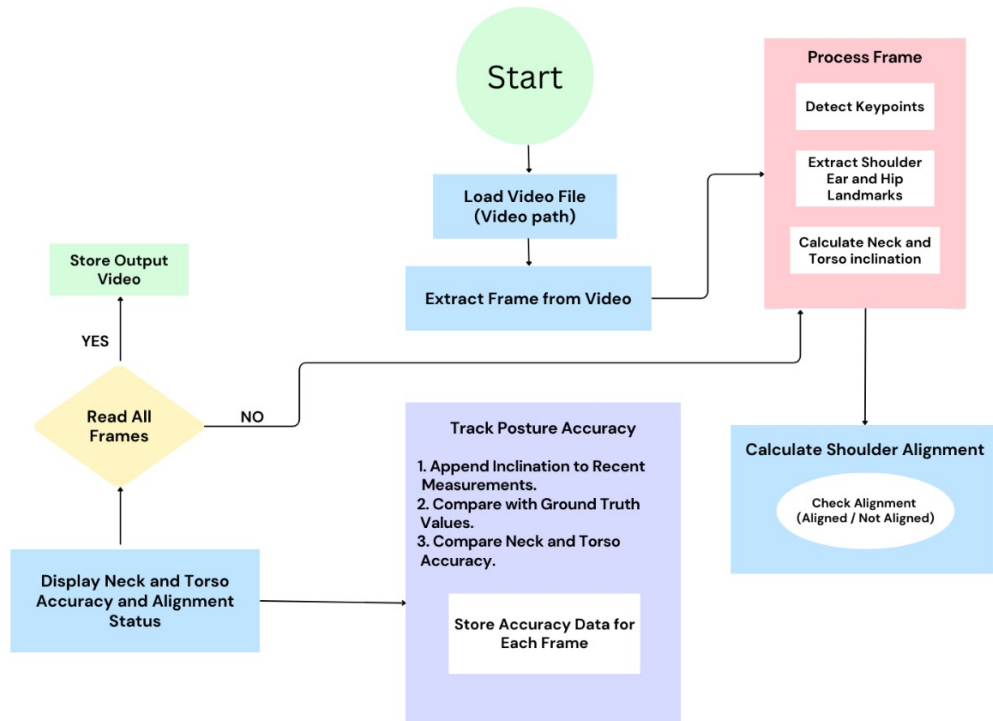


Fig. 1: Flowchart of the posture detection system: A visual representation detailing the process from video capture, keypoint detection, and angle calculation to the generation of real-time feedback.

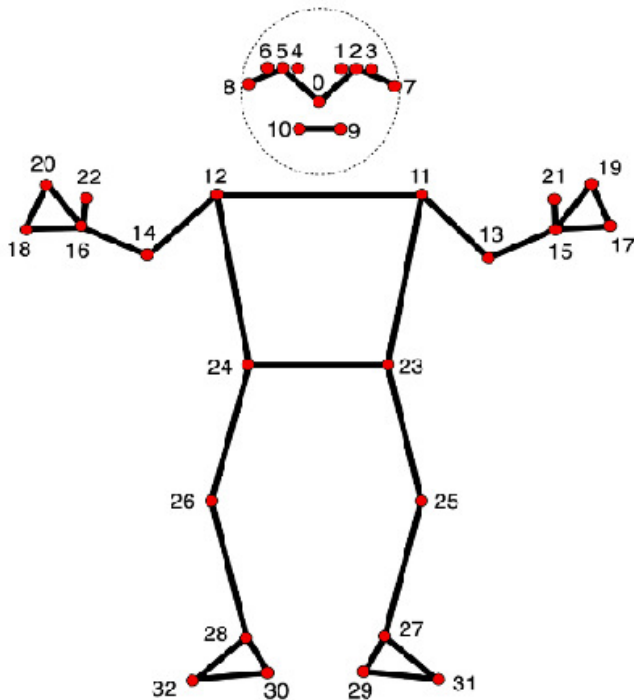


Fig. 2: Mediapipe body keypoints used in posture analysis. Key points include neck, shoulders, and hips, which are essential for assessing sitting posture.

F. Conditions for Correct Posture

The system classifies posture as "good" or "bad" based on angle measurements and predefined thresholds.

- Neck Inclination: Acceptable within 10-15 degrees relative to torso. Deviation away from this range indicates forward head posture.
- Torso Inclination: The deviations permit being within a certain range to maintain an erect posture, but variations of greater than 5-10 degrees are interpreted as slouching
- Shoulder Offset: For appropriate posture, the vertical separation between the shoulders has to be at least minimal; bigger deviations require greater correction. Such a requirement can vary with the size of the input frame.

G. Real-Time Feedback Mechanism

Instant feedback from this system is achieved by implementing a GUI at each point. The instant the angles reach the tolerance limits, the GUI sends a visual signal to alert the users to change the posture. The feedback mechanism is relentless in that it updates every frame passing through the system based on the video stream; thus, advisory support is timely.

H. Experimental Setup

The programming language used for the system implementation is Python, paired with Mediapipe for keypoint detection, and OpenCV for video processing. The experimentation was done on a standard desktop with a 2.5 GHz processor, 16GB RAM, and an operating webcam at 30 frames per second. The experimentation includes testing this system in terms of accuracy in posture deviation detection and responsiveness for real-time monitoring.

IV. RESULTS AND DISCUSSION

A. Linear Improvement Analysis:

The system could be described as showing an upward trend in accuracy but in a very initial phase. Interestingly, for epochs 1-10, the system depicts a linear progression in terms of accuracy improvement, which essentially means the model has learned and adapted very well. But after crossing this threshold, the trajectory levels off at such a point that the model has stabilised at a certain performance level.

B. Gaussian Distribution Fit Analysis:

1) *Neck Angle Accuracy Distribution:* The maximum accuracy of neck angle measurements reached 92%. The average accuracy is about 92.23% as shown by the dashed line in the histogram. Despite the Gaussian distribution used on the dataset, the fit is not perfect. That means there are minor deviations from a normal distribution. These can be attributed to the limits of the dataset, which may introduce variability in terms of measurement efficacy and detection.

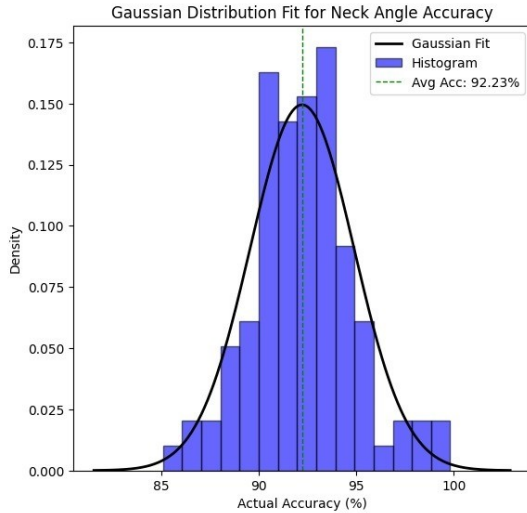


Fig. 3: Neck Angle Analysis Results

2) *Torso Angle Accuracy Distribution:* Therefore, the torso angle accuracy reached its maximum value up to 90% and on average, it has reached 90.08%, as shown by the dashed line. In comparison with the neck angles, torso angles data had more symmetry and better overlap with the Gaussian fit and therefore presented fewer variability in the dataset performance. Results mean that the system has been better

in reality in stable recognition of the torso postures than the neck postures.

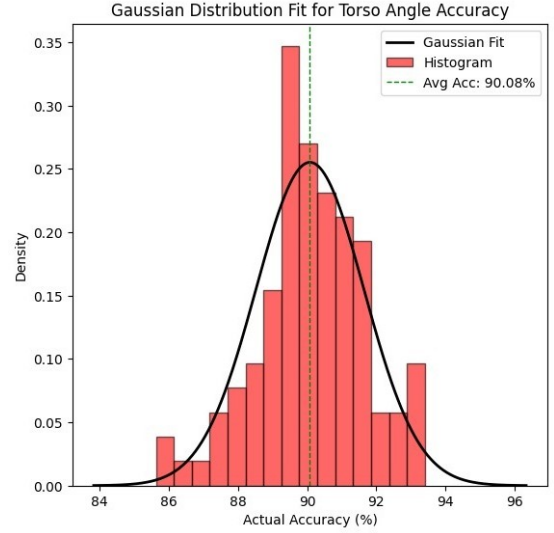


Fig. 4: Torso Angle Analysis Results

These analyses depict the strength and weaknesses with respect to posture recognition. Linear improvement shows that the learning is done correctly; however, analysis through the Gaussian fit also provides information about distribution and also which direction needs more improvement. Fig. 3 and Fig.4 displays the results of our analysis that tend to follow a Gaussian curve.

The angle-based sitting posture recognition system was tested with a sample video and the following performance metrics were obtained :

1. Accuracy: 84.41% - The total percentage of correctly identified poses, which reflects the overall reliability of the system in classification.
2. Precision: 99.00% - High precision demonstrates that the system has a low false positive rate, accurately detecting correct postures without misclassification.
3. Recall: 70.41% - The recall rate reflects the system's ability to identify all instances of correct posture, with some room for improvement in ensuring complete detection.
4. F1 Score: 82.63% - The harmonic mean of precision and recall, highlighting a balanced performance that favors high precision while still considering recall.

Additional images for results

C. Comparison with Existing Methods

When compared to other vision-based models like OpenPose or DeepPose, which typically achieve around 90-93% accuracy but require higher computational power, the proposed system offers a more resource-efficient solution. It achieves competitive accuracy with a simpler, angle-based approach that allows for real-time feedback even on standard desktop setups. Additionally, the dynamic tolerance mechanism of the proposed system adapts to individual variations, providing more personalized feedback.

The distribution analyses of neck and torso angles provide further insights into the system's learning capability. The relative symmetry in the torso angle accuracy suggests effective detection of stable body regions, while the variability in neck angle accuracy indicates potential areas for enhancement, such as increasing training data for more consistent results. Overall, these metrics and comparisons illustrate that the proposed system strikes a balance between precision, adaptability, and real-time performance, making it suitable for practical applications like workplace posture correction.

D. Challenges and Limitations

The major challenges the development phase faced included diversity of data and real-time processing. Diverse body types, clothes, lighting, and conditions of backgrounds affected the system's performance making need a diverse dataset to make it reliable. Real-time processing requires efficient algorithms and huge computing power to ensure that it delivers a very high frame rate where posture detection is done with no latency.

It suffers from limitations of failing to make distinctions between the slightest postural variations such as tilts versus slumping, and susceptibility to false positives caused by occlusions or camera angle changes. It also suffers from limited data sets, with risks of overfitting and generalization to new unseen data being suboptimal. More data and improved algorithms will reduce these problems.

V. CONCLUSION

The Sitting Posture Recognition System has immense potential in the computer vision and machine learning framework to detect different sitting postures, thereby making it very practical for promoting ergonomic practices in work and learning places. The system is strongly effective in detecting standard postures and could be potentially very useful in real applications. However, at the moment, it is relatively weak with slight posture variations as well as adaptation to other conditions in the environment. These challenges depict how there is scope for data diversity and algorithm refinement in order to be accurate as well as flexible.

Conclusively, angle-based sitting posture recognition was proven to be effective by analyzing neck and torso angles, with torso showing higher accuracy and fit to the Gaussian distribution pattern. The system is thus very ideal for long-term real-time monitoring applications because of its adaptive learning capability. Neck posture measurements are generally skewed to a lesser extent, which can be attributed to a lesser training dataset; increasing the same will further enhance accuracy and contribute towards valuable insights for posture correction and overall health promotion.

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